



“The pilot line is important. Once the chips are qualified, it becomes easy for commercial plants to start production. This is a major milestone”

ASHWINI VAISHNAW

Union Minister for
Railways, I&B,
and Electronics & IT

tion equipment in Coimbatore, shipping its first unit to Micron in Gujarat with 40% local sourcing, cutting costs by 30%, says Manickam. Deposition equipment applies thin layers of material to surfaces. While YES declined to share customer names or exact savings, it confirmed supplying global customers from its Sullur plant.

Vaishnaw, too, points at clustering in Sanand. “If you look around this place, you will find very large tanks, which are basically the chemical guys. If you see, neighbouring plants manufacture substrates, and there are plants which are now manufacturing the wire bond and other stuff which goes into semiconductors,” he says. Recently, a delegation of 57 Japanese companies visited these sites and more or less decided to set up units here, Vaishnaw told *BT* at the inauguration of the CG Semi pilot line. CG Semi says it still depends on imported equipment and materials but sees a strong domestic momentum building up.

Besides, semiconductor manufacturing is as much about people as it is about equipment and materials. India is struggling despite Indians accounting for 20% of the global chip design workforce. The government has launched a mission to train 85,000 engineers for semiconductor manufacturing. AICTE has introduced new curricula such as B.Tech in Electronics (VLSI Design & Technology) and Diploma in IC Manufacturing. MeitY has provided EDA (electronic design automation) tools to 280 institutes and 72 start-ups. But much of the expertise lies in industry practices, process optimisation, and operational know-how evolved over decades. “It is essential to go beyond classroom learning. India must collaborate with leading global semiconductor firms and

New Beginnin

2020

- ▶ The government launches the Scheme for Promotion of Manufacturing of Electronic Components and Semiconductors, with a 25% incentive on capital expenditure for semiconductors, ATMP (assembly, testing, marketing, packaging), and other critical components
- ▶ Swadeshi Microprocessor Challenge to encourage homegrown microprocessor development, resulting in prototypes such as SHAKTI (IIT Madras) and VEGA (C-DAC) being taken up by start-ups and academia for IoT, automotive, and defence applications

2021

- ▶ ₹76,000 crore Semicon India Mission launched. Popularly called Semicon 1.0

2025

- ▶ **May:** Cabinet approves HCL-Foxconn JV for advanced chip packing
- ▶ **August:** Four More Projects approved —SiCSem, to establish integrated facility of Silicon Carbide-based Compound

Semiconductors in Odisha
—Continental Device India, to expand its discrete semiconductor manufacturing facility in Punjab
—3D Glass Solutions, to vertically integrate advanced packaging and embedded glass substrate

unit in Odisha
—Advanced System in Package Technologies’ OSAT in Tamil Nadu
—CG Power’s end-to-end pilot line inaugurated
▶ **September:** —India’s first indigenously developed Vikram-32-bit chip presented to the prime minister at the Semicon India 2025 conference. Semicon 2.0 awaited